Bitwise Operators in C - Practice Questions

Question 1: Ticket Counter

Problem Statement

A cinema gives odd-numbered tickets to Gate A and even-numbered tickets to Gate B. Write a C program that accepts a ticket number and decides which gate it goes to using the bitwise AND operator (`n & 1`). If the last bit is 1, the number is odd, otherwise it is even.

Input Format

A single integer N, the ticket number (1 <= N <= 10000).

Output Format

Print Gate A if N is odd, otherwise print Gate B.

Sample Input	Sample Output
`10`	`Gate B`

Test Cases

Input	Output
`7`	`Gate A`
`100`	`Gate B`

Question 2: Permission Checker

Problem Statement

A file system stores a user's permissions in one number, with one permission per bit. Bit positions are counted from the right, starting at 0. Write a C program that checks whether the permission at position K is enabled, using `(n >> k) & 1`.

Input Format

Two integers N and K separated by a space (1 <= N <= 10000, 0 <= K <= 13).

Output Format

Print ON if the K-th bit is 1, otherwise print OFF.

Sample Input	Sample Output
`10 1`	`ON`

Test Cases

Input	Output
`10 0`	`OFF`
`10 3`	`ON`

Question 3: Enable a Feature

Problem Statement

A mobile app stores its enabled features in one number, with one feature per bit. The admin wants to turn ON the feature at position K, leaving all other features unchanged. Write a C program that sets the K-th bit to 1 using `n | (1 << k)` and prints the updated number.

Input Format

Two integers N and K separated by a space (1 <= N <= 10000, 0 <= K <= 13).

Output Format

Print the new value of N.

Sample Input	Sample Output
`10 2`	`14`

Test Cases

Input	Output
`10 0`	`11`
`1 3`	`9`

Question 4: Disable an Alarm

Problem Statement

A security system stores the state of its alarms in one number, with one alarm per bit (1 = active, 0 = disabled). The guard wants to switch OFF the alarm at position K without touching the others. Write a C program that clears the K-th bit using `n & ~(1 << k)` and prints the new state.

Input Format

Two integers N and K separated by a space (1 <= N <= 10000, 0 <= K <= 13).

Output Format

Print the new value of N.

Sample Input	Sample Output
`10 1`	`8`

Test Cases

Input	Output
`15 3`	`7`
`255 7`	`127`

Question 5: Smart Fan Button

Problem Statement

A smart home panel stores the state of its devices in one number, with one device per bit. Pressing a button flips the device at position K: ON becomes OFF, and OFF becomes ON. Write a C program that toggles the K-th bit using `n ^ (1 << k)` and prints the new state.

Input Format

Two integers N and K separated by a space (1 <= N <= 10000, 0 <= K <= 13).

Output Format

Print the new value of N.

Sample Input	Sample Output
`12 2`	`8`

Test Cases

Input	Output
`7 0`	`6`
`8 1`	`10`